Fruit Pulper



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Fruit pulper

What it is used for

A fruit pulper separates edible pulp from seeds, skins, and other coarse material using a rotating beater and screen. Correct fruit preparation, screen selection, and hygienic handling help maximise pulp recovery and maintain product quality.

Main parts

- Fruit feed hopper
- Rotating beater or pulping rotor
- Perforated screen or sieve
- Motor and drive assembly
- Pulp outlet and residue discharge outlet
- Protective guards, control panel, and emergency stop

Before you start

1. Confirm the fruit type, required pulp texture, production capacity, and intended downstream process.
2. Inspect the hopper, rotor, screen, seals, outlets, guards, and electrical connections for damage.
3. Ensure all food-contact parts are clean, sanitised, and assembled according to the manufacturer's instructions.
4. Check that the pulp and residue collection containers are clean, food-grade, and correctly positioned.
5. Wash fruit and remove unsuitable foreign material, hard stones, metal, or damaged portions according to the approved preparation method.
6. Verify that the screen is suitable for the fruit and that the emergency stop and guards function correctly.

How to use

1. Switch off and isolate power before installing a screen, opening the pulper, or adjusting the rotor assembly.
2. Fit the approved screen and verify that it is correctly seated and secured.
3. Close the covers, secure the guards, and ensure that both discharge outlets are clear.
4. Position separate clean containers for pulp and rejected residue.
5. Start the machine using the approved control and allow it to reach normal operating speed.
6. Feed prepared fruit gradually and evenly through the hopper without exceeding the rated capacity.

7. Use only the manufacturer's approved feed pusher where supplied; never reach into the hopper during operation.
8. Monitor pulp flow, residue discharge, motor load, vibration, and any unusual noise.
9. Check that pulp consistency and recovery meet the process requirements and that the screen is not blocked.
10. Stop the machine if the rotor jams, the screen shifts, the motor overloads, or product leaks from an unintended location.
11. Isolate power and wait for all moving parts to stop before clearing a blockage or removing internal material.
12. Collect the pulp in clean, covered containers and transfer it promptly to the next processing stage.
13. Inspect the pulp for unwanted seeds, skin fragments, or foreign material according to the quality procedure.
14. At the end of processing, stop and isolate the machine before disassembly and cleaning.
15. Clean and sanitise the screen, rotor, hopper, and outlets using the approved procedure.

⚠ Safety

- Only trained personnel should operate the fruit pulper.
- Keep hands and tools away from the rotor, screen, and feed mechanism while the machine is running.
- Never bypass guards or operate the machine with loose covers.
- Isolate power before clearing jams, changing screens, or performing maintenance.
- Do not force oversized fruit or hard materials into the pulper.
- Handle sharp screens and heavy removable components using appropriate PPE and safe lifting methods.
- Prevent cleaning water from entering electrical parts unless the machine is designed for the approved washdown procedure.

Care & storage

- Clean and sanitise the hopper, screen, rotor, seals, and discharge outlets after each production run.
- Inspect the screen for holes, wear, deformation, or blockage.
- Check rotor blades or beaters, bearings, seals, and fasteners for damage or looseness.
- Keep the motor and electrical controls dry and clean ventilation openings regularly.
- Arrange servicing if pulp recovery declines, the screen becomes damaged, or the machine vibrates excessively.

Common mistakes to avoid

- Using an unsuitable screen that produces poor pulp quality or low recovery.
- Feeding fruit too quickly and overloading the rotor.
- Allowing residue or pulp to remain in the machine and cause contamination.
- Clearing a blocked screen before stopping and isolating the machine.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Prepared by - Subodh Kumar